

PICL Rally Timing — Requirements

Sep 23, 2026 · @Someone

PICL needs a low-cost way for volunteers at up to 8 timing points to record each rider's plate and start/finish times on their own phones, working without signal, so rally results can be published accurately and defended if challenged. This doc states what the system must do; the architecture is left to the build team.

Problem and goal

A rally event times riders across several timed segments, and each segment has its own start and finish point. At a recent PICL rally, manual (likely pen-and-paper) recording across these points produced results that couldn't be trusted, and they were scrapped. The goal is results that are complete, accurate and auditable, published the same day, at near-zero cost per event.

Success looks like:

- Every rider who completes a segment has a recorded start and finish, or is explicitly flagged.
- Any disputed time can be checked against evidence captured at the moment of crossing.
- Results flow into piclstats.com without anyone re-keying them.
- Volunteers with no training beyond a short briefing can run a station.

Users and roles

Role	Who	What they need to do
Station volunteer	Parent or coach volunteer, one or two per timing point	Record plate and time for each rider at one start or finish point, on their own phone
Timing lead	One PICL official or designated volunteer	Set up the event and stations, monitor station status, resolve issues, approve results

Role	Who	What they need to do
Race official	PICL staff	Review disputes and view evidence for a specific rider and segment
Public	Riders, families, coaches	View published results on piclstats.com

Functional requirements

The minimum viable version for a first live trial is everything marked Must. Should and Could items can follow.

ID	Requirement	Priority
Event setup		
F1	The timing lead can create an event with its segments, and each segment has one start point and one finish point (up to 8 points in total, possibly more).	Must
F2	The timing lead can load the rider roster (plate, name, team, category), ideally from the existing registration export.	Must
F3	A volunteer can join a specific station in one step (for example by scanning a printed code) without creating an account.	Must
F4	Once joined, the device clearly shows which event, segment and point (start or finish) it is recording for.	Must
Time capture		
F5	A volunteer can record a crossing with a single action, at the moment it happens, without first entering the plate.	Must
F6	A volunteer can record several riders crossing within a second or two of each other without losing any of them.	Must

ID	Requirement	Priority
F7	At a start point, a volunteer can alternatively enter the plate first and then record the start.	Should
F8	Each recorded time has at least one-second precision. Tenths are preferred.	Must
Rider identification		
F9	A volunteer can attach a plate number to any recorded crossing after it was recorded.	Must
F10	Plate entry validates against the roster and warns on unknown plates and on a plate already recorded at this point.	Must
F11	The system suggests the plate automatically from the photo, and the volunteer confirms it.	Could
Evidence		
F12	Each crossing can capture a photo taken at the moment of recording, stored with that record.	Should
F13	A station can record continuous video of its line as a backup, and an official can find the frame that matches a disputed crossing.	Could
F14	Officials can view the evidence for any rider and segment.	Should
Corrections and audit		
F15	Volunteers and the timing lead can correct a record: change its plate, void it, or add a missed crossing with a note.	Must
F16	No record is ever silently changed or deleted. Every correction keeps the original, who made the change, when, and why.	Must
F17	A station can keep a manual backup (paper sheet or equivalent) that the timing lead can enter afterwards, flagged as manually entered.	Must
Sync and transfer		

ID	Requirement	Priority
F18	Records sync to a central store automatically whenever the device has a connection, without volunteer action.	Must
F19	Each station shows how many records are waiting to sync and when it last synced.	Must
F20	Records can be transferred from a station to the timing lead's device with no internet at all.	Should
F21	Any station can export its records as a file as a last resort.	Must
F22	Syncing the same record more than once never creates a duplicate.	Must
Reconciliation and results		
F23	The system pairs each rider's start and finish for each segment and calculates segment times and the rally total.	Must
F24	The system flags: a start with no finish, a finish with no start, duplicate plates, unknown plates, negative or implausible times, and stations that haven't synced.	Must
F25	The timing lead can resolve each flag, with evidence visible alongside it where captured.	Must
F26	The timing lead can view provisional results during the event.	Should
F27	The timing lead approves results before they are published, and results can't be published while flags are unresolved unless they are explicitly overridden.	Must
F28	Approved results appear on piclstats.com in the same form as other race results.	Must
F29	Results can be exported in a format PICL can use for its official scoring.	Must

ID	Requirement	Priority
Access		
F30	Only people with a station code or timing-lead access can create or change records. The timing pages are not linked from the public site.	Must

Non-functional requirements

ID	Area	Requirement
N1	Offline	Every station function except sync works with no signal for the whole event. No record is lost if the app is closed, the phone locks, or the battery dies and the phone restarts.
N2	Timing accuracy	A segment time is a finish time minus a start time recorded on two different phones. That calculated time must be accurate to within 1 second. The system must detect and report any device whose clock can't be trusted to that level.
N3	Devices	Works on volunteers' own recent iPhones and Android phones with no app store install, or a single install at most.
N4	Field usability	Usable one-handed, outdoors in bright sun, with gloves or wet hands. The main action is operable in under a second, and a new volunteer is productive after a 5-minute briefing.
N5	Battery	A station can run for 4+ hours on a typical phone battery, with photo capture on.
N6	Scale	Supports at least 8 concurrent stations, 300 riders and 2,500 crossings per event without slowing down.
N7	Auditability	The full history of every record, including corrections and manual entries, is kept and can be exported after the event.

ID	Area	Requirement
N8	Cost	Running costs are near zero per event. No per-event or per-rider licence fees, and no dedicated timing hardware beyond what volunteers already carry.
N9	Privacy	Riders are minors. Rider names and photos are visible only to authorised officials, photos are kept only for a defined retention period, and public results show only what PICL already publishes.
N10	Results turnaround	Provisional results are available within 30 minutes of the last rider finishing, given at least one device with a connection.

Constraints and out of scope

Constraints:

- It lives inside piclstats.com and is hidden from the main navigation until PICL adopts it.
- Venues such as Birdsboro Preserve may have little or no mobile signal at timing points.
- Volunteers change from race to race, so there is no training beyond a race-day briefing.
- It is built and run by a volunteer, so the solution needs little ongoing maintenance.

Out of scope for now:

- Dedicated timing hardware such as RFID chips, timing mats or transponders.
- Mass-start and cross-country race formats, which PICL already times.
- Rider registration and payment.
- Live public leaderboards during the event.

Open questions

For PICL:

- ☐ How is a rally scored: the sum of segment times, or a placing per segment? Do transfer sections count, and is there a cutoff time?
 - ☐ Sum of segment times. No cutoff. Transfers are not timed.

- ☐ What happens when a rider misses a segment or has no time for it: a DNF, a penalty time, or something else?
 - ☐ TBD. I don't think it's possible to miss a segment as the course is a single lap and coaches are all through the course. If someone has a mechanical/injury, they will miss latter segments and count as a DNF.
- ☐ Can riders repeat a segment, and if so which run counts?
 - ☐ No - just one run
- ☐ What format does PICL need to import results into its official scoring?
 - ☐ csv
- ☐ What is PICL's policy on photographing riders, who are minors, and how long can photos be kept?
 - ☐ There is a release form signed by parents
- ☐ How many volunteers can realistically be placed at each point?
 - ☐ There are 3 total. Check https://docs.google.com/document/d/1ibYEKMJD_bBYhl-3pn9VF6v12iwn1w3kfnwnsddefPw/edit?tab=t.0 for details. The captain at each segment start/finish is responsible for timekeeping.

For the architect:

- ☐ How will N2 timing accuracy be achieved and verified across devices that may never get a signal during the event?
- ☐ Can photo or video capture meet N5 on battery life, or does it need to be optional per station?
- ☐ What is the preferred no-internet transfer method for F20?
- ☐ Should we compare the build against an off-the-shelf multi-device timing app (for example Webscorer) on cost and fit before committing?